# AE 358 RENEWABLE ENERGY SYSTEMS AND MANAGEMENT

## LECTURE 5

## BIODIESEL

Lecturer: Francis Kemausuor

# COURSE STUDY GUIDE

| Date | Meeting/ Week # | Topic for the day | Activity |
|---|---|---|---|
| 14-16 Jan | 1 | | Lecture |
| 19-23 Jan | 2 | Introduction to Energy Systems | Lecture |
| 26-30 Jan | | Renewable Energy Systems and their Potential | |
| 2-6 Feb | 3 | Bioethanol | Lecture |
| 9-13 Feb | 4 | Hydropower / Group assignments | Lecture |
| **16-20 Feb** | **5** | **Biodiesel** | **Lecture** |
| 23-27 Feb | 6 | Biogas | Lecture |
| 2-6 Mar | 7 | | Lecture |
| 9-13 Mar | 8 | Mid-semester exams | Quiz |
| 16-20 Mar | 9 | Discuss group assignment challenges | Lecture |
| 23-27 Mar | 10 | Solar Energy | Lecture |
| 30 – 3 Apr | 11 | Wind Energy | Lecture |
| 6-10 Apr | 12 | Renewable Energy Applications in Agriculture | Lecture |
| 13-17 Apr | 13 | GROUP PRESENTATIONS | Student-led |
| 13-17 April | 14 | GROUP PRESENTATIONS | Student-led |

# Housekeeping

- Thanks for attending the Saturday lecture
- How was the experience?
- Where are my valentine gifts?

# Lecture outline

- **Feedstocks for biodiesel production**
- **Technical aspects of biodiesel production**
- **Uses of biodiesel**
- **Economics of biodiesel**

# Biodiesel: production, use and economics

# What is biodiesel?

- Biodiesel is usually referred to as a mixture of fatty acid methyl esters (FAME) produced from vegetable oils and animal fats via transesterification reaction.
- It has been also defined as the monoalkyl esters of the long chain fatty acids derived from renewable feedstocks, such as vegetable oils or animal fats, for use in compression ignition (diesel) engines.

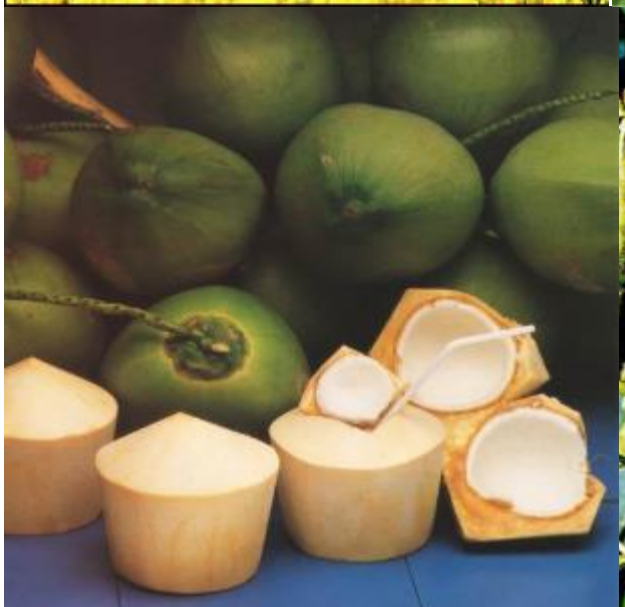
# Biodiesel feedstocks



- Oils from the following sources:
  - Rapeseed
  - Soybean
  - Jatropha
  - Oil palm
  - Cottonseed
  - Sunflower
  - Coconut
  - Microalgae
  - Animal fats
  - waste oil
  - Etc.









# Jatropha



# Biodiesel production

- Oil Extraction
- Oil Refining
- Transesterification





# Oil extraction

- **Industrial:** centralized production by refining in large industrial plants. E.g. Juaben Oil Mills

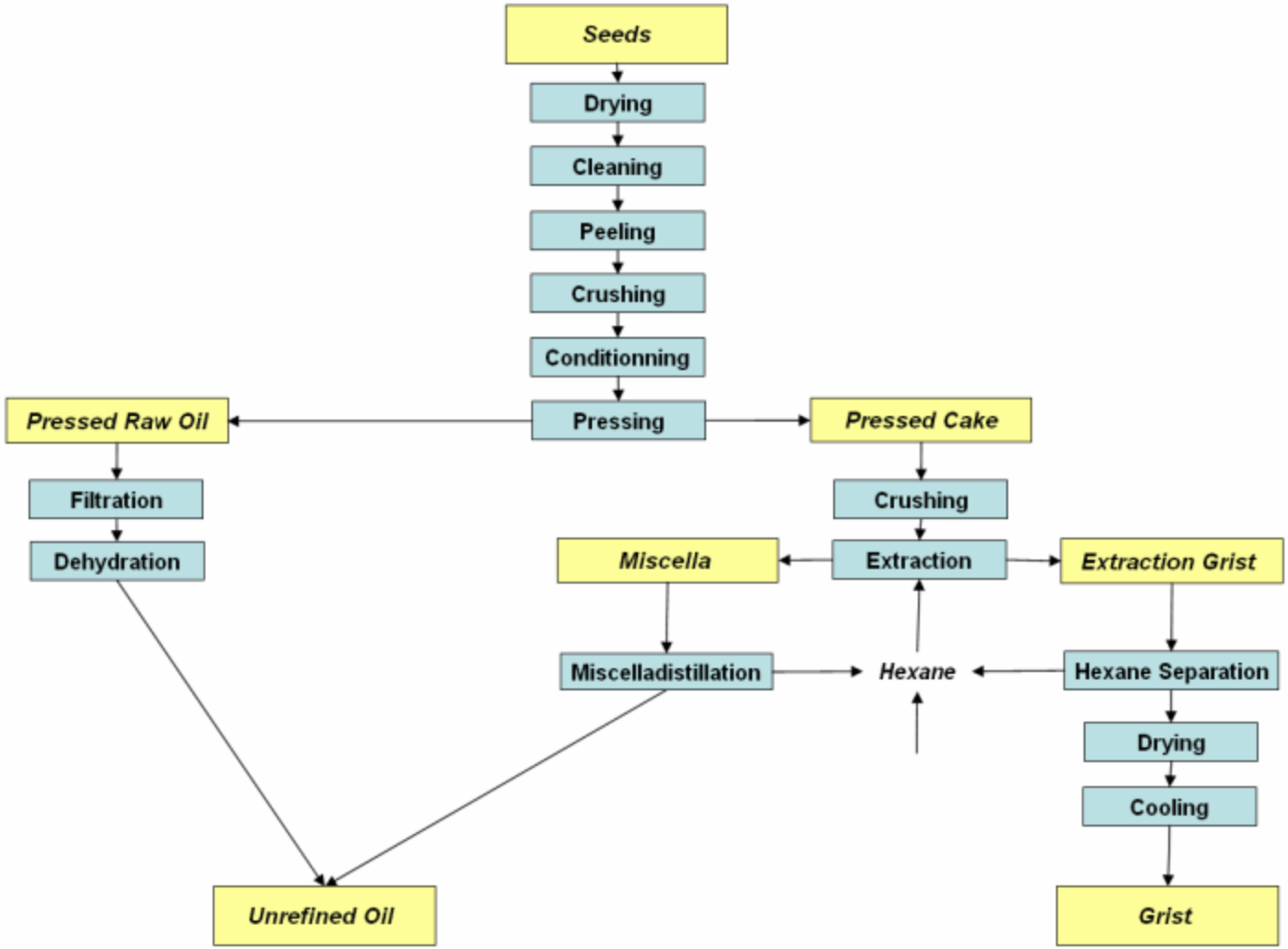
- **Small scale pressing:** decentralized cold pressing directly on farms or in cooperatives



# Small scale pressing

- The cleaned oil seeds are exclusively mechanically pressed at maximum temperatures of 40 ºC.
- Suspended solids are removed by filtration or sedimentation.
- The press cake (co-product) is left with a remaining oil content of usually over 10 %, which is used as a protein-rich fodder (non-poisonous crops)
- Due to higher production costs, the decentralized oil production by farmers is not widely applied today, although there is the chance of additional income for farmers

# Industrial processing

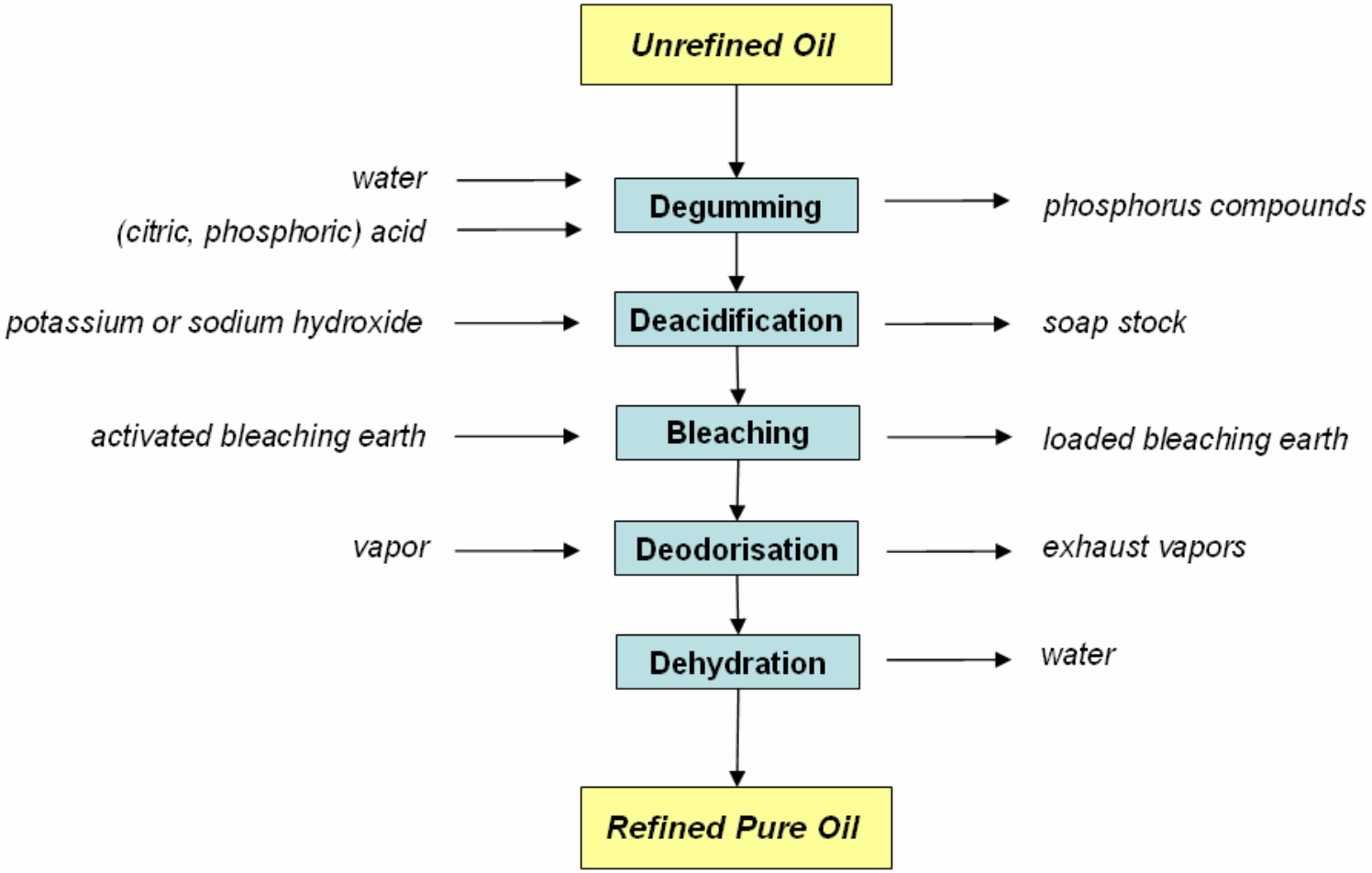


# Oil refining

- Refining is important in order to remove undesirable substances, such as phosphatides, free fatty acids, waxes, and colorants.
- These substances can alter oil storage life and hamper further processing.
- During this first refining step the oil mass (4 to 8 %) and the solvent contents are reduced.
- The refining process depends on the vegetable oil quality, and the refining steps depend on the feedstock source.



# Oil refining process



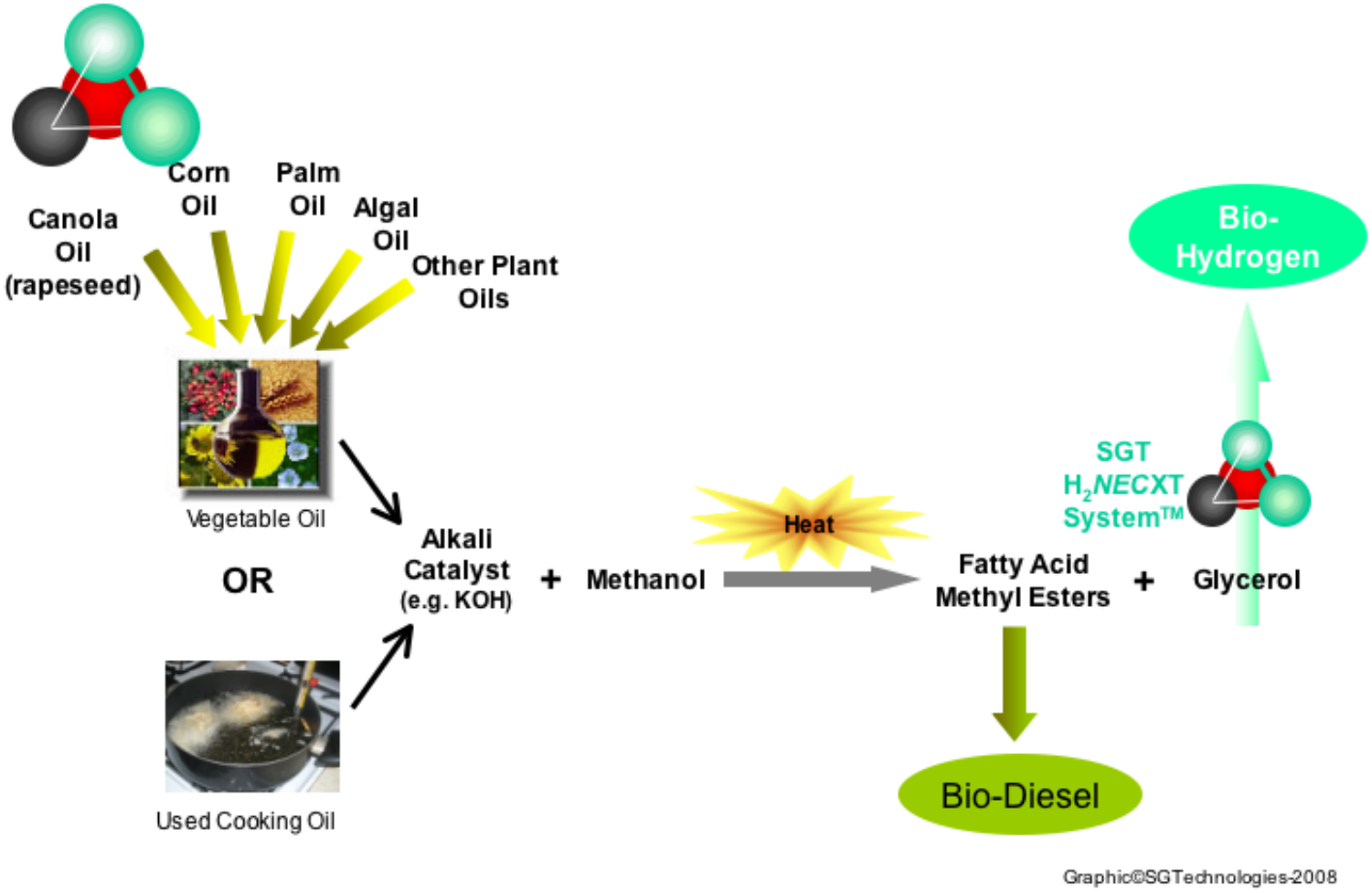
# Oil refining process

- The first purification step of oil refining is the removal of phosphatides, also known as **degumming.** This is necessary as phosphatides make the oil become turbid during storage and as they promote the accumulation of water
- The second refining step is the **deacidification.** There are several methods for achieving this but ***Neutralization with alkali*** is the most applied method. FFA‟s are saponified with alkaline solutions and the resulting soap is separated.

# Oil refining process

- In the third step **bleaching,** colorants are removed. This process step enhances storage life of the biofuel.
- In the **deodorization** step odorous substances are removed by steam distillation.
- Finally a **dehydration** step has to be conducted, as traces of water may decrease conversion in the transesterification process of biodiesel production. The removal of water is either accomplished by distillation or by passing a stream of nitrogen through the fatty material

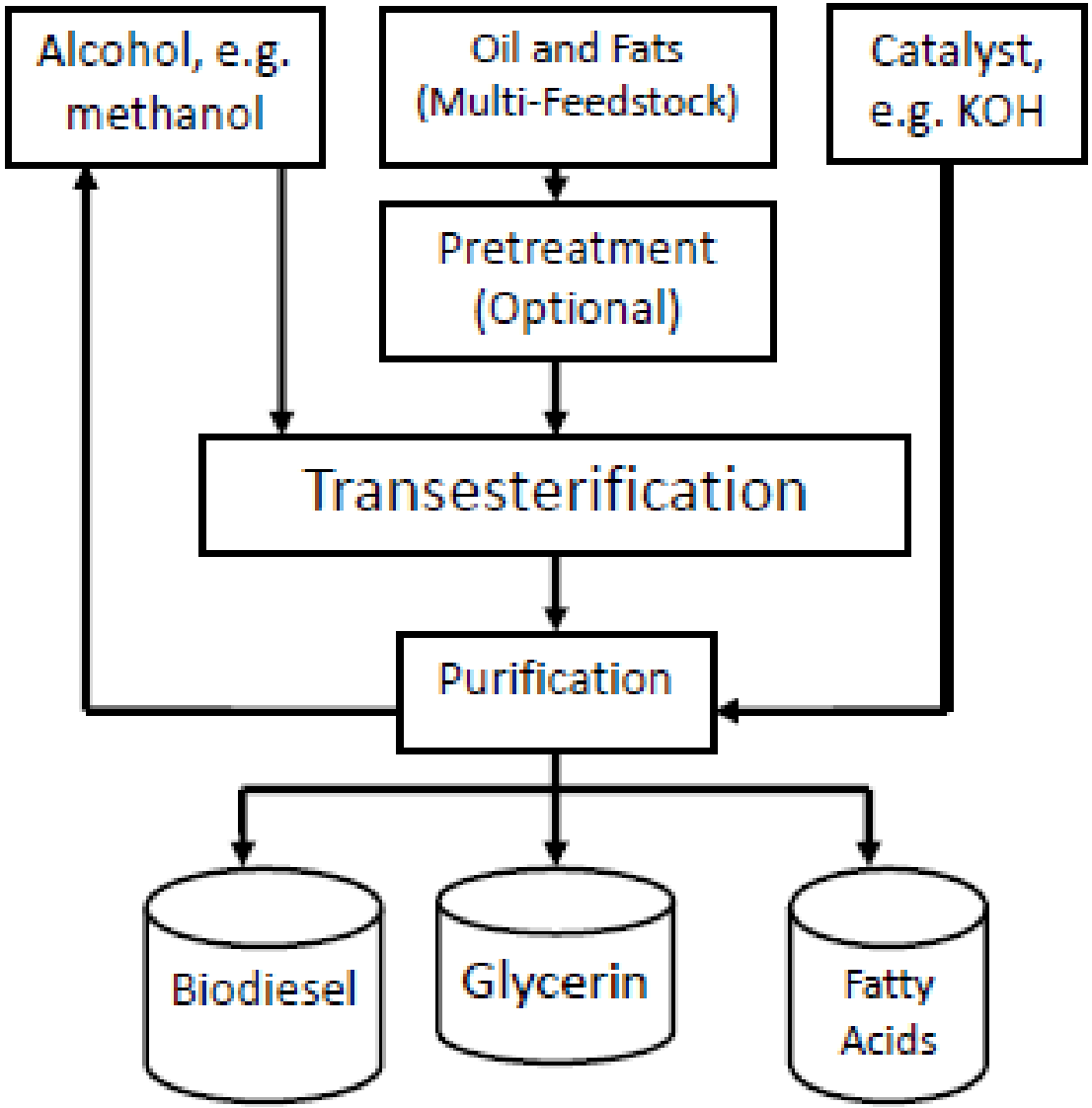
# TRANSESTERIFICATION

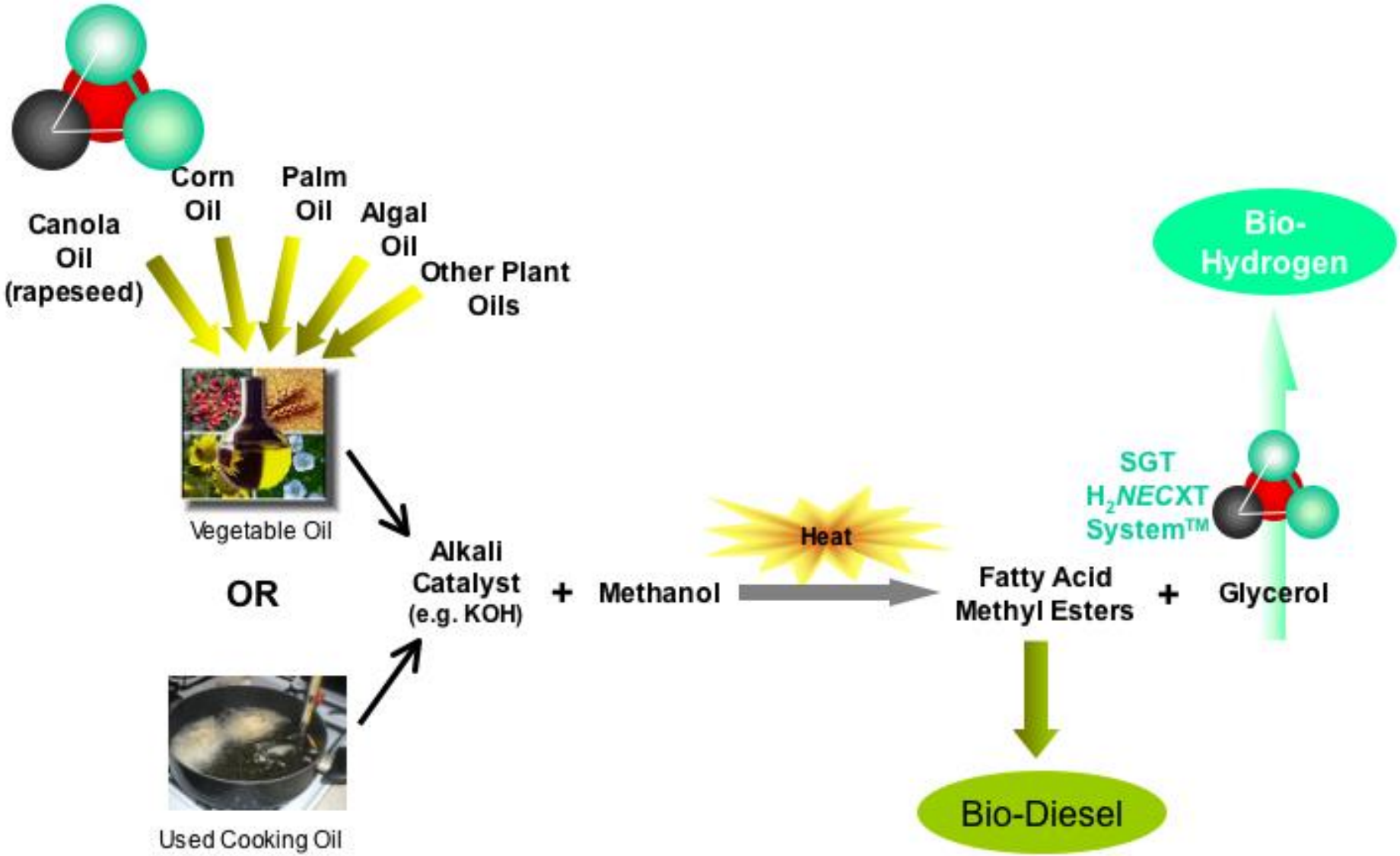


# Why transesterification?

- Straight vegetable oils (SVO) could be used as motor fuel. The major problem however, is that SVO has high viscosity which cause compression ignition problem in diesel engines. E.g. leaving deposits on various engine parts.
- To overcome these drawbacks, several techniques have been implemented. These include: blending with fossil diesel, transesterification, micro-emulsification and thermal cracking or pyrolysis for the production of biodiesel from SVO.
- Among all these production techniques the transesterification of oils has seen wider application due to its easy, cost effective process technology and also because the final products meets physico-chemical parameters of the fuel quality.

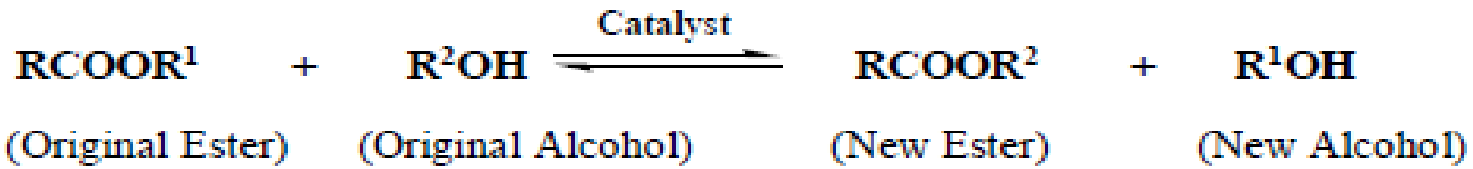
# The transesterification reaction

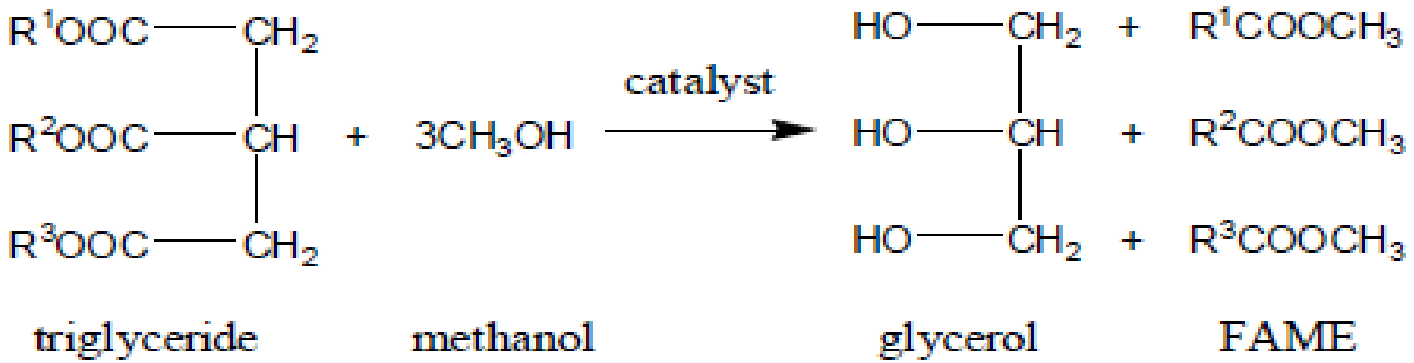



Canola Oil (rapeseed)
Corn Oil
Palm Oil
Algal Oil
Other Plant Oils
Vegetable Oil
OR
Used Cooking Oil
Alkali Catalyst (e.g. KOH)
+
Methanol
Heat
Fatty Acid Methyl Esters
+
Glycerol
SGT H2NECXT System™
Bio-Hydrogen
Bio-Diesel

# Transesterification reaction

- Transesterification is the chemical conversion of oil to its corresponding fatty ester (biodiesel). The process involves reacting an animal fat or vegetable oil (triglycerides) with an alcohol, in the presence of an acid or alkali catalyst to produce an alcohol ester.



# Factors that influence the Transesterification Reaction

1. Types and amount of catalysts.
2. Quality of feedstocks such as free fatty acids, moisture content, etc.
3. Types and molar ratio of alcohol to triglycerides, and
4. Temperature, reaction time and mixing intensity of the reaction mixture.

# Catalysts

- Although the transesterification process could proceed in the absence of catalysts, the reaction usually is conducted by using catalysts due to economic reasons. Non-catalytic reactions, even though create purer esters and soap-free glycerin, take place too slowly and high energy inputs are required. Several catalysts can be used for the transesterification process.
  - Alkaline material
  - Acidic material
  - Transition metal compounds
  - Silicates
  - Lipases

# Properties of biodiesel

- Generally, the properties of biodiesel and especially its viscosity and ignition properties are similar to the properties of fossil diesel.
- The energy content per litre of biodiesel is about 5 to 12 % lower than that of diesel fuel
- Biodiesel however has several advantages. For example the cetane number and lubricating effect of biodiesel, important in avoiding wear to the engine, are significantly higher.

# Properties of Biodiesel

- Additionally, the alcohol component of biodiesel contains oxygen, which helps to complete the combustion of the fuel. The effects are reduced air pollutants such as particulates, carbon monoxide, and hydrocarbons.
- Since biodiesel contains practically no sulfur, it can help reducing emissions of sulfur oxides.

# Biodiesel blends

- Biodiesel mixes easily and completely with fossil diesel at any concentration. Typical blends are B5, B20, and B30 with 5%, 20%, and 30% biodiesel content respectively.
- B100 (100% biodiesel) offers the most environmental benefits. Use of B100 may however require engine or fuel system component modification and can cause operating problems, especially in cold weather.
- B20 (20% biodiesel) offers about one fifth of the environmental benefits of B100, but can be more broadly applied to existing engines with little or no modification.
- B2 (2% biodiesel) offers little environmental or petroleum dependence benefit and could be potentially used as an environmental marketing tool.

# Applications of Biodiesel



# Compression Ignition Engines

- Biodiesel can be used in compression ignition engines either blended with fossil diesel or unblended (B100).
- Most diesel vehicles are able to run on blends of up to B20 with few or no modifications, particularly if the vehicle was manufactured after the mid-1990s. For older models susceptible plastic and rubber components must be replaced by more resistant materials.
- While the use of low blends requires no or only minor technology modifications, the use of higher blends, such as B100, need more efforts. It may require modification of engine or fuel system components as well as some fine tuning.

# Application in Agriculture

- Biodiesel can be used to run agricultural machinery. In rural agricultural communities, biodiesel programmes can help provide the energy needs for their farming activities if they mobilize themselves into cooperatives to own the farm machinery. Such an arrangement can also help to increase their income.
- A major challenge in the production and use of biodiesel is the use of food crops, e.g., palm oil, maize, soya beans, sorghum, and coconut oil for the production of fuel. This challenge can be overcome if non-edible crops are employed. *Jatropha curcas*, which grows quite well on marginal lands and is toxic to human beings, can serve as an energy crop to fill that gap.